# Geometric Sequence in Assembly

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### Assembly Source Code

Yigal Hoffner Mano CPU Simulator version 17 16 Oct 2010

Paste your code into this window (Ctrl-V)

```
ORG 0
cla       // clear AC
lda Q     // load Q value to AC
cma       // complement AC
inc       // double increment (-Q)
sta Q_M   // store AC value at Q_M
lda A1    // load A1 value to AC
sta SAVE_LOC I // store AC value at the address stored in SAVE_LOC
sta N_TERM     // store AC value at N_TERM
bsa OUT_subroutine   // jump to OUT_subroutine & save return adress
hlt            // halt the CPU

OUT_subroutine, DEC 0         // saving the return point of the OUT_subroutine
LOOP_OUT, cla                 // clear AC
          lda Q_M             // load Q_M value to AC
          sta COUNT_Q         // store AC value at COUNT_Q
          lda N_TERM          // load N_TERM value to AC
          sta L_TERM          // store AC value at L_TERM
          isz SAVE_LOC        // increment SAVE_LOC value & skip if zero
          bsa IN_subroutine   // jump to IN_subroutine & save return adress
          sta N_TERM          // store AC value at N_TERM
          sta SAVE_LOC I      // store AC value at the address stored in SAVE_LOC
          isz COUNT_9         // increment COUNT_9 value & skip if zero
          bun LOOP_OUT        // branch unconditionally to LOOP_OUT
          bun OUT_subroutine I  // return from OUT_subroutine

IN_subroutine, DEC 0          // saving the return point of the IN_subroutine
               cla            // clear AC
LOOP_IN, add L_TERM           // add L_TERM value to AC
         isz COUNT_Q          // increment COUNT_Q value & skip if zero
         bun LOOP_IN          // branch unconditionally to LOOP_IN
         bun IN_subroutine I  // return from IN_subroutine

ORG 100           // start of data section at 100h
A1, DEC 2         // the first term of the sequence
Q, DEC 3          // the common ratio (Q) of the sequence
Q_M, DEC 0        // number of iterations of the IN_LOOP
COUNT_Q, DEC 0    // holds Q_M for repeated multiplication routine
COUNT_9, DEC -9   // number of iterations of the OUT_LOOP
N_TERM, DEC 0     // The resulting(new) term after multiplication
L_TERM, DEC 0     // the last term that needs to be multiplied by itself Q times
SAVE_LOC, HEX 200 // stores the location of the array elements
ORG 200       // start of data section at 200h
aa1, DEC 0    // array elements
a2, DEC 0
a3, DEC 0
a4, DEC 0
a5, DEC 0
a6, DEC 0
a7, DEC 0
a8, DEC 0
a9, DEC 0
a10, DEC 0
end           // end code
```

Assemble/Load | Clear | Cancel

**Variables & Subroutines**

**Main Variables:**

- A1: The first term of the geometric sequence.
- Q: The common ratio of the sequence.
- Q_M: Stores the negative value of Q (-q) to be used as a loop counter.
- COUNT_9: Counter for the outer loop (runs 9 times to generate the next 9 elements).
- COUNT_Q: Counter for the inner multiplication loop.
- N_TERM / L_TERM: Temporarily hold the newly calculated term and the previous term during the multiplication process.
- SAVE_LOC: A memory pointer (starting at address 200h) where the sequence elements are saved.

**Subroutines:**

- OUT_subroutine (Outer Loop): Manages the overall sequence generation. It runs 9 times, calls the inner subroutine to calculate the next number, and stores each result in the memory array.
- IN_subroutine (Inner Loop): Since basic Assembly lacks a multiplication command, this subroutine performs multiplication using repeated addition. It adds the previous term to the Accumulator Q times.

## Execution Results (Memory State)




